
Artificial Intelligence Fundamentals

Search, evolutionary optimization and artificial life, logical and probabilistic agents

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1 Introduction

Artificial intelligence (AI) is a large family of methods: machine learning (ML) is one member, deep learning (DL) is a branch of ML, and the large language models (LLMs) behind tools such as ChatGPT and Copilot are a branch of DL. This course covers “all AI, but learning”, in the lecturer’s words: methods that answer broader questions than ML and that run, mostly unseen, inside many AI-based tools. Some learning may still appear; Section 1.7 explains the choice.

After a lecture on the design of intelligent agents (Section 2), the course runs in four blocks:

- **Solving problems by search:** uninformed and informed search, local search, constraint satisfaction problems, search in games.
- **Evolutionary optimization and artificial life:** evolutionary optimization, novelty-based search and quality-diversity, artificial life.
- **Logical agents:** inference in propositional and first-order logic, planning.
- **Probabilistic agents:** Bayesian networks, hidden Markov models.

Multi-agent systems close the course.

1.1 Symbolic AI

Symbolic AI, the dominant paradigm from the 1950s to the 1990s, represents the world at a high level, in a form that humans can read and interpret. Its building block is the symbol.

Definition 1.1 (Symbol). A **symbol** is a meaningful pattern that can be manipulated.

A symbol stands for something else, as the string `cat` stands for the animal, and a program can compare, copy and combine it into rules. Semiotics, the study of signs, calls the pattern the **signifier** and what it stands for the **signified**.

Where do the symbols of a machine come from? Either humans provide them, and then they mean something only through someone who already had symbols: a chicken-and-egg problem, which the lecturer relates to Hofstadter’s “strange loops”. Or the computer builds its own low-level representations from raw data, which is sub-symbolic manipulation (Section 1.5).

1.2 The symbol grounding problem

With the first answer, the machine manipulates symbols without knowing what they stand for. Harnad named this difficulty.

Definition 1.2 (Symbol grounding problem). The **symbol grounding problem** asks how a symbol is connected to the real-world object it stands for: what makes the connection, who makes it, and how. In Harnad’s words: “How can the semantic interpretation of a formal symbol system be made intrinsic to the system, rather than just parasitic on the meanings in our heads?”¹

¹S. Harnad, “The symbol grounding problem”, *Physica D* 42 (1990), pp. 335–346.

A symbol represents something outside the system, so the answer requires “jumping out of the system”, in Hofstadter’s phrase. Formal logic meets the same limit in Gödel’s incompleteness theorem: every consistent, effectively axiomatized system that can express elementary arithmetic has a true statement that it cannot prove, and establishing that statement takes a step outside.

1.3 The physical symbol system hypothesis

If symbols can be grounded, is manipulating them enough for intelligence? Newell and Simon answered yes, as a hypothesis.

Definition 1.3 (Physical symbol system). A **physical symbol system** is a system that manipulates symbols to produce other symbols. It is *physical* because the manipulation is a physical process, which consumes energy.

In the lecture’s terms, humans are physical symbol systems, and so are digital computers, since both manipulate symbols. The **physical symbol system hypothesis** (PSSH) states:

A physical symbol system has the necessary and sufficient means for general intelligent action.²

Newell and Simon gloss both halves. *Necessary*: “any system that exhibits general intelligence will prove upon analysis to be a physical symbol system”. *Sufficient*: “any physical symbol system of sufficient size can be organized further to exhibit general intelligence”. The claim is empirical, so it is a hypothesis and not a theorem: evidence can support or refute it. If it holds, a suitably organized digital computer can act with general intelligence.

1.4 Intelligence is not consciousness

Intelligence and consciousness are different properties: a system that acts intelligently need not be conscious. Consciousness is usually tied to subjective experience: in Nagel’s sense,³ a being is conscious if there is something it is like to be that being. Such experience is hard to detect from outside, even by dissecting a system and studying its behavior. A **zombie**, which behaves exactly like a person but has no experience, would pass every behavioral test. Searle’s Chinese room makes a related point about understanding: a person who follows written rules to manipulate Chinese characters answers questions in Chinese without understanding any.⁴

If consciousness exists, materialism holds that it is a by-product of the body, located “somewhere” in it; dualism instead treats the mind as a substance different from the body. Hofstadter argues that consciousness depends on the symbolic pattern, whatever its physical substrate: silicon chips organized in an equivalent way could support it just as neurons do. The lecture leaves the question open.

²A. Newell and H. A. Simon, “Computer science as empirical inquiry: symbols and search”, *Communications of the ACM* 19(3) (1976), pp. 113–126.

³T. Nagel, “What is it like to be a bat?”, *The Philosophical Review* 83(4) (1974).

⁴J. R. Searle, “Minds, brains, and programs”, *Behavioral and Brain Sciences* 3(3) (1980).

1.5 Sub-symbolic representations

Symbolic descriptions break down when the world varies more than rules can anticipate; the alternative lets the machine build its own representation.

Definition 1.4 (Sub-symbolic representation). A **sub-symbolic representation** encodes a concept as a numeric pattern spread over many low-level units, such as the weights of a neural network, instead of as a single symbol that a human can read.

Example 1.5 (Recognizing a cat). This example contrasts the two approaches. A symbolic system describes a cat through symbols for its parts, such as two eyes, whiskers and pointed ears, and through rules that combine them. Real photographs of cats vary in pose, breed, lighting and framing, and each new variation breaks some rule. A sub-symbolic system learns its representation from many example images and absorbs the variation, but no human can read that representation.

1.6 Models of the world

An AI system reasons about a **model of the world**: a representation of the part of reality that it works on. The lecture distinguishes three kinds, by the question that each answers.

- **Descriptive**: how is the world? A map of Pisa.
- **Predictive**: what value will an unobserved quantity take? The selling price of a house from its floor area, number of rooms and energy efficiency; machine learning builds such models.
- **Decisional**: what should be done to reach a goal? The path through a labyrinth; the search methods of the course answer such questions.

1.7 Hybrid systems

Machine learning maps inputs X to outputs Y . Building the map (training) needs huge amounts of data, more for more complex mappings; using it (inference) needs expensive memory, time and accelerators such as GPUs and TPUs. Deep learning needs even more of both. It still works very well in many applications, and the lecture states the course's position explicitly: "This course is not saying 'don't use deep learning'. It's saying 'don't use deep learning only'." A **hybrid system** combines a learned model with components that are not learned.

Example 1.6 (Retrieval-augmented generation). Retrieval-augmented generation (RAG) is a worked case of a hybrid system. A **retriever** fetches the documents relevant to a question from an external memory, such as Wikipedia, and a **generator**, a language model, writes the answer from the question and those documents. We can extend the memory to a new domain or update it with the news without retraining, and each answer can cite its source document, which makes it easier to interpret.

2 Intelligent agents

The course builds AI systems as agents: systems that perceive an environment and act on it. This section fixes the vocabulary that the rest of the course uses.

2.1 Agents and environments

To reason about any AI system we first separate the part that decides from the world it decides about.

Definition 2.1 (Agent). An **agent** is anything that can act. It perceives its **environment** through **sensors** and acts on it through **actuators**.

Where the agent ends and the environment begins is the designer's choice, but it must be fixed before we can study the agent. What the sensors deliver at one instant is a **percept**. Time runs in discrete steps $t = 1, 2, \dots$, one percept per step, so the agent's experience is a sequence of percepts.

Definition 2.2 (Agent function). The **agent function**, or **policy**, is a map

$$f: P^* \rightarrow A,$$

where P^* is the set of percept histories, the finite sequences of percepts, and A is the set of the agent's actions.

In words: f says which action the agent takes after every possible percept history. We can implement it physically, in a robot, or simulate it in a program.

Example 2.3 (Vacuum-cleaner world). This worked example runs through the section. A vacuum cleaner moves between two squares, A and B. Its actions are $A = \{Left, Right, Suck, NoOp\}$ and its position is $x \in \{A, B\}$. The status of the square that it occupies, $s_t \in S = \{clean, dirty\}$, is known only through perception, and the percept at time t is the pair (x_t, s_t) . The task is to clean the environment.

Building an agent means building f . Algorithm 1, a first attempt, chooses from the current percept alone. It learns nothing and contains no AI, because the designer wrote every case by hand.

2.2 Rationality

Algorithm 1 makes sense intuitively, but calling it a *good* agent needs a criterion. Most definitions of AI ask the agent to do “the right thing” and never mention intelligence: this narrows the field, but it makes the goal precise enough to build agents toward it.

Moral philosophy offers several notions of the right thing. AI adopts **consequentialism**: we evaluate an agent by the consequences of its actions, that is, by the sequence of environment states that its behavior produces; its intentions and its own opinion of its success do not count. The choice suits a machine, because consequences can be observed and measured and intentions

Algorithm 1 Reflex vacuum agent

Require: the percept (*location, status*)**Ensure:** an action

```

1: function REFLEX-VACUUM-AGENT(location, status)
2:   if status = dirty then
3:     return Suck
4:   else if location = A then
5:     return Right
6:   else if location = B then
7:     return Left
8:   end if
9: end function

```

cannot. It also costs work, because we must say which consequences are good. In a simple world we could list them; in a complex one no complete evaluation of every behavior is possible, so we summarize the good consequences in a single number.

Definition 2.4 (Performance measure). A **performance measure** evaluates the goodness of an environment state, or of a sequence of environment states.

Definition 2.5 (Rational agent). For each possible percept sequence, a **rational agent** selects an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.

In words: a rational agent does the best it can with what it has perceived and what it knew from the start, where “best” is fixed by the performance measure. That single number is where the difficulty lies, as Goodhart’s law warns: “when a measure becomes a target, it ceases to be a good measure”.

Example 2.6 (A measure that backfires). This example shows a measure failing. We cannot enter the room to check it, so we reward the vacuum cleaner with the grams of dirt that it picks up over a one-hour shift. A maximizing agent can dump the dirt back and pick it up again: the score grows and the room is never cleaner. The measure rewarded a behavior that we expected to be useful, instead of the environment state that we wanted.

Rationality also asks less than perfection. A rational agent maximizes *expected* performance given the percept history, so:

- **rational is not omniscient:** the percepts may not supply all the relevant information;
- **rational is not clairvoyant:** the agent has no access to future percepts;
- **rational is not successful:** the outcome of an action may not be the expected one.

In every case, careful engineering of the performance measure, and control of the behavior that the agent learns, are key.

2.3 Fixed action patterns

An environment that is completely observable and predictable never changes its optimal solution, so it encourages the agent to learn one fixed behavior. The result is a fragile agent that follows a **fixed action pattern**: it keeps executing its policy even when the world changes under it.

Example 2.7 (The Sphex wasp). Nature provides a contrast case. After digging a burrow, the Sphex wasp stings a caterpillar, drags it to the burrow, checks inside that all is well, drags the caterpillar in and lays its eggs. If an entomologist moves the caterpillar a few inches while the wasp is checking, the wasp returns to the “drag the caterpillar” step and checks the burrow again, unchanged, even after dozens of such interventions.

2.4 Task environments: PEAS

Before designing an agent we describe its task. A **PEAS** description lists the **P**erformance measure, the **E**nvironment, the **A**ctuators and the **S**ensors; Table 2.1 gives one for an autonomous taxi. The actuators and the sensors form the **agent architecture**, and the agent is the architecture together with the agent function.

Table 2.1. PEAS description of an autonomous taxi service.

Element	Autonomous taxi
Performance measure	safety, reaching the destination, profits, legality, comfort, ...
Environment	streets and freeways, traffic, pedestrians, weather, ...
Actuators	steering, accelerator, brake, horn, speaker or display, ...
Sensors	cameras, accelerometers, gauges, engine sensors, keyboard, GPS, ...

The more restricted the environment, the easier the design of the agent.

2.5 Properties of environments

Which agent to build depends on the environment, so we classify the environment first: four dimensions, plus a fifth on what the agent knows.

- **Observable.** The environment is **fully observable** if the sensors give access to its complete state, **partially observable** if faulty sensors or missing information hide part of it, and **unobservable** if the agent has no sensor information at all.
- **Deterministic.** The environment is **deterministic** if the current state and the action uniquely determine the next state, and **nondeterministic** (or stochastic) if the same state and action can lead to different next states. Partial observability is one possible cause of nondeterminism.
- **Discrete.** The state of the environment, time, the percepts and the actions can each take **discrete** or **continuous** values.
- **Single agent.** The environment is **multi-agent** if the performance of our agent depends on the actions of other agents, and **single-agent** otherwise.

- **Known.** In a **known** environment the agent knows its rules, that is, which next state follows from each state and action. A known environment can still be partially observable. In an **unknown** environment the agent may try to learn the rules.

Table 2.2 classifies three environments. The real world sits in the hardest column: like the taxi, it is partially observable, nondeterministic, continuous and multi-agent.

Table 2.2. Three environments along the four dimensions.

Characteristic	Sudoku	Backgammon	Taxi service
Fully observable	yes	yes	no
Deterministic	yes	no	no
Discrete	yes	yes	no
Single-agent	yes	no	no

2.6 Types of agent

Agents differ in how they compute the agent function. We go from the simplest design to the most flexible one; each design adds a component that removes a limit of the previous one.

2.6.1 Table-driven agents

The most direct way to build f is to write it down. A **table-driven agent** stores its whole percept history and looks up the action in a table indexed by percept sequences (Algorithm 2).

Algorithm 2 Table-driven agent

Require: a percept

Ensure: an action

persistent: *percepts*, a sequence, initially empty; *table*, a table of actions indexed by percept sequences, initially fully specified

- 1: **function** TABLE-DRIVEN-AGENT(*percept*)
 - 2: append *percept* to the end of *percepts*
 - 3: *action* $\leftarrow$ LOOKUP(*percepts*, *table*)
 - 4: **return** *action*
 - 5: **end function**
-

The design is only feasible in simple environments. With a set P of possible percepts, the table needs one entry per percept sequence, $\sum_{t=1}^T |P|^t$ entries for an agent that lives T steps: far too many once the environment and the agent grow complex. The design trades efficiency for effectiveness.

2.6.2 Simple reflex agents

A **simple reflex agent** chooses its action from the current percept alone, through **condition-action rules**: if the percept satisfies a condition, then the agent takes the corresponding action. The name comes from the body: like a reflex, the agent reacts to the stimulus without consulting any memory. Algorithm 1 is a simple reflex agent.

The simplification cuts the cost dramatically: the table grows exponentially with the lifetime T , while a simple reflex agent needs one rule per percept and work linear in T . It simplifies an already naive approach, though, and it only works in small, fully observable environments, which are rare.

2.6.3 Model-based agents

A simple reflex agent fails as soon as the current percept does not reveal the state. A **model-based agent** keeps an **internal state**, which depends on the percept history, and uses it to estimate the current state of the environment. Two models update the internal state:

- the **transition model** describes how the world evolves, and in particular the consequences of the agent's actions;
- the **sensor model** describes how the state of the world shows up in the agent's percepts.

By approximating the entire state, a model-based agent handles partially observable environments well. Acquiring such a model is hard: the lecture notes that large language models still struggle to acquire one.⁵

2.6.4 Goal-based agents

Knowing the state is not enough to know what to do. A **goal-based agent** also represents **goals**, descriptions of the states it wants to reach, and it predicts what the world will be like if it takes an action. To choose, it must weigh the immediate reward of an action against the expected long-term reward. Two families of methods explore the options: **search**, which navigates the tree of possible decisions, and **planning**, which exploits knowledge about the world to execute the best actions.

Goals are what a reflex agent lacks: its condition–action rules model a single problem, so a new goal means rebuilding it from scratch, while a goal-based agent represents its goals explicitly and can change them.

2.6.5 Utility-based agents

A goal only separates success from failure. A **utility-based agent** assigns a value, its **utility**, to every intermediate state or sequence of states, not only to the final goal: utility works as an internal performance measure. In stochastic, partially observable environments, where utility-based agents are common, the agent chooses the action with the highest **expected utility** over its possible results i :

$$EU = \sum_i p_i u_i,$$

where p_i is the probability of result i and u_i is its utility.

2.6.6 Learning agents

Every design so far is fixed by its designer. A **learning agent** improves itself through four components:

- the **policy** chooses an action, given the context (the percepts and the internal state);

⁵The slide points to M. Mitchell, “LLMs and world models, part 1”.

- the **critic** evaluates the current behavior against an external performance measure and sends feedback;
- the **learning element** uses the feedback to change the other components;
- the **problem generator** steers the agent away from greedy behavior by trying out new actions, which balances the critic.

Learning is orthogonal to the previous types: any of them can be a learning agent or not, and learning can modify any of its components.

2.7 Knowledge representation

Every agent above represents the world somehow. The lecture places a representation on two axes.

Expressiveness. An **atomic** representation treats each state as indivisible, with no internal structure. A **factored** representation describes a state as a vector of attribute values. A **structured** representation describes a state as a graph: objects with their properties and the relationships among them. Expressiveness grows from atomic to structured.

Distribution. In a **localist** representation, concepts and memory locations correspond one to one. In a **distributed** representation, one memory cell takes part in several concepts, and one concept is spread over several cells.